

Parth Mital

📍 Vadodara, Gujarat, India ✉ parth.mital.2004@gmail.com 🔗 linkedin.com/in/parthmital
🐙 github.com/parthmital 🌐 parthmital-portfolio.vercel.app/

Education

Vellore Institute of Technology (VIT)

B.Tech in Computer Science & Engineering (CGPA: 8.70 / 10)

2023 – 2027

Vellore, India

Projects

🔗 NeuroScan AI - Medical Imaging Research & Full-Stack Analysis Platform

- Benchmarked **21 deep learning architectures** on **9,000+ MRI scans** for automated brain tumour analysis.
- Detection: 99.59% accuracy** using CoaTNet-0, outperforming ResNet/Inception baselines.
- Classification: 98.58% accuracy**; reduced dataset noise by **19%** via perceptual hashing **deduplication**.
- 3D Segmentation: 3D U-Net** achieved Dice scores of **0.847 (WT)**, **0.753 (TC)**, **0.761 (ET)**.
- Built **clinical SaaS platform** with **JWT authentication**, **NIFTI pipelines**, & **vtk.js 3D visualization**.
- Tech:** Python, PyTorch, TensorFlow, FastAPI, React 18, TypeScript, SQLModel, Tailwind, vtk.js, OpenCV.

🔗 The Obsidian Flask - AI Game Engine with Memory-Driven NPCs

- Developed an **AI narrative engine** where NPCs **speak naturally**, adapt to choices, and evolve over time.
- Implemented **Retrieval-Augmented Generation (RAG)** system enabling NPCs to recall **1000+ interactions**.
- Created **deterministic LLM orchestration** using **LangGraph** for structured and contextually consistent dialogue.
- Built **real-time event-sourced backend** supporting **50+ concurrent sessions** with live updates.
- Tech Stack:** Python, FastAPI, Pydantic, SQLite, LangGraph, Groq API, RAG, FAISS, Sentence Transformers, React, TypeScript, Vite, Tailwind CSS, Zustand, React Router, TanStack Query, WebSockets, Docker, Uvicorn.

🔗 VIT Carpool - Full-Stack Campus Ride-Sharing Platform | 🌐 vitcarpool.vercel.app

- Built **centralized ride-sharing platform** replacing fragmented WhatsApp-based coordination.
- Implemented **structured filtering system** (source, destination, time) for **real-time** efficient ride discovery.
- Achieved **2,600+ views/impressions** within the VIT student community, indicating strong engagement.
- Secured access via domain-restricted **Google OAuth** & enforced data isolation using **Supabase RLS**.
- Tech:** React 18, TypeScript, Vite, Tailwind CSS, Supabase, PostgreSQL, REST APIs, Google OAuth.

🔗 VIT Resource Automation Pipeline - Academic Resource Lifecycle Toolkit

- Built an **end-to-end automation pipeline** processing **6,000+ academic documents** per semester, reducing collection time from **weeks** to under **1 hour**, including asynchronous Selenium/Playwright scrapers for **PYQs**.
- Automated **bulk data extraction** from VTOP/UniBud across **150+ faculty** & **6 concurrent subjects** and implemented an **analysis engine** (pypdf + COM) to rank faculty by slide count & content depth.
- Standardized** legacy files (.doc, .ppt) into 100% PDF format for consistent access.
- Tech:** Python, Selenium, Playwright, pypdf, pywin32, comtypes, Chrome DevTools Protocol, CSV.

Technical Skills

Languages: C, C++, Java, Python, TypeScript, JavaScript

Relevant Coursework: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Operating Systems, Computer Networks, Digital Systems Design, Microprocessors & Microcontrollers

Software Development: Full-Stack Development, Frontend, Backend, Databases, REST APIs, Authentication

Machine Learning & Data: Machine Learning, Computer Vision, ETL Pipelines, Data Processing, Automation

Cloud & DevOps: Containerization, Cloud Deployment, AWS, Docker

Experience

Hiraeth Tech

Web Development Intern

Jun 2025 – Jul 2025

Vadodara, India

- Built **5+ responsive WordPress websites** using Elementor & custom CSS.
- Optimized images & assets to improve page load performance.
- Ensured cross-device responsiveness & consistent UI/UX.